Bootstrapping language-guided navigation learning with self-refining data flywheel

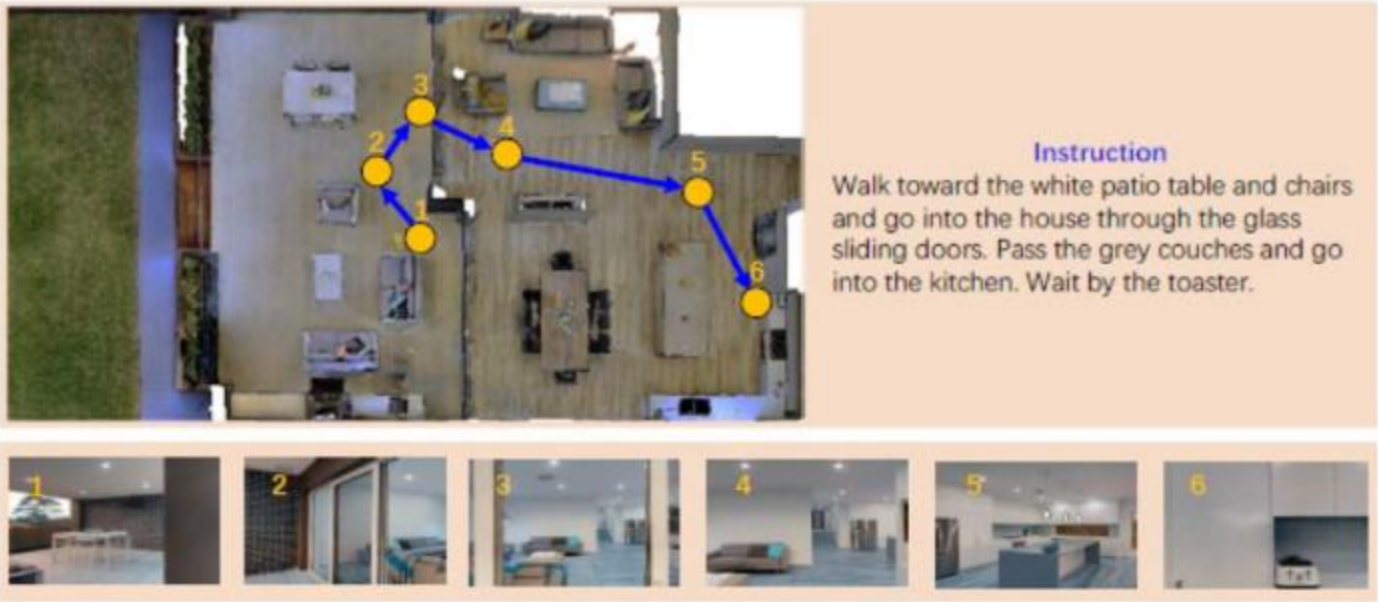
Published as a conference paper at ICLR 2025

BOOTSTRAPPING LANGUAGE-GUIDED NAVIGATION LEARNING WITH SELF-REFINING DATA FLYWHEEL

Zun Wang[1,2*] **Jialu Li**[2] **Yicong Hong**[3] **Songze Li**[1] **Kunchang Li**[1] **Shoubin Yu**[2]
Yi Wang[1] **Yu Qiao**[1] **Yali Wang**[1] **Mohit Bansal**[2] **Limin Wang**[1,4]
[1]Shanghai AI Laboratory [2]UNC Chapel Hill [3]Adobe Research [4]Nanjing University
{zunwang, jialuli, mbansal}@cs.unc.edu, wanglimin@pjlab.org.cn

2025/10/17

Background

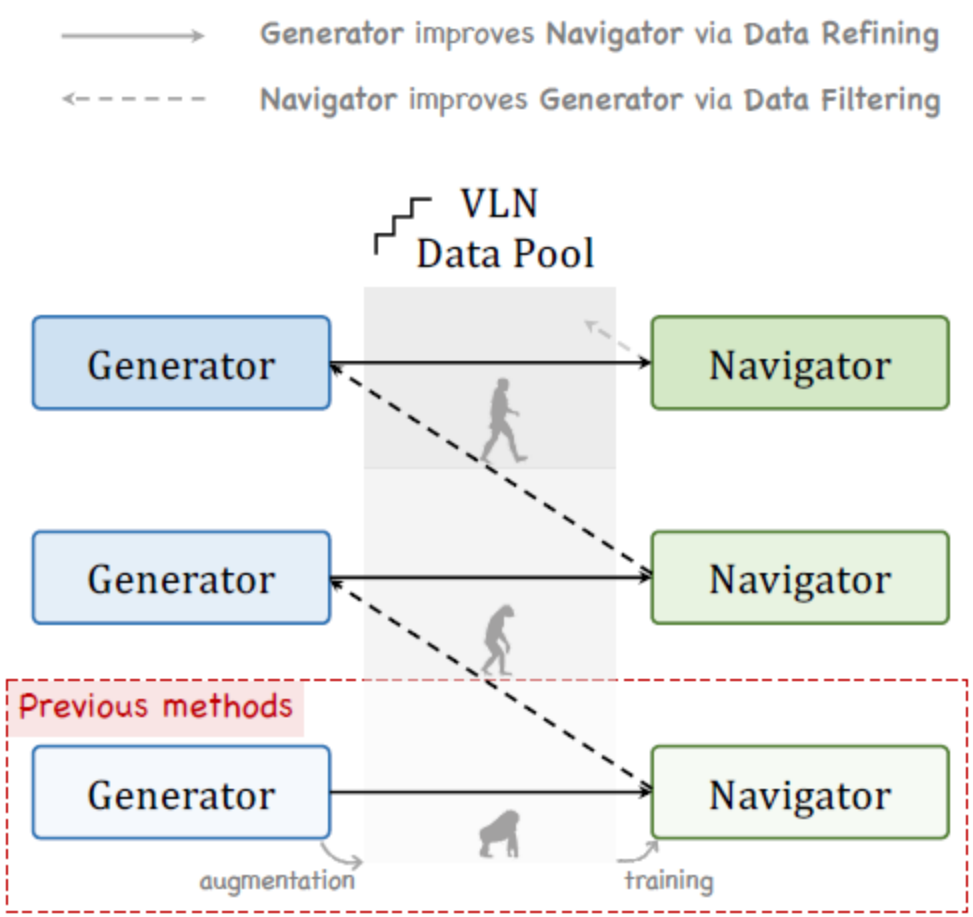


- 在常见的视觉语言导航(VLN)任务中，智能体接收一条自然语言指令，通过不断观察周围环境并执行动作，移动到指定位置。

- 环境被建模为一个无向拓扑图$\mathcal{G}=\{\mathcal{V},\mathcal{E}\}$，其中$\mathcal{V}$表示可导航的节点，$\mathcal{E}$表示连接边。

- 任务开始时被放置于未见过环境中的一个起始节点。在每一个导航的时间步t，智能体接收到当前所处节点$\mathcal{V}_t$的全局视觉观察，全景观察包含36个独立的视角图像$\mathrm{R}_t=\{\mathrm{r}_{t,\mathrm{i}}\}_{\mathrm{i}=1}^{36}$，而后智能体根据导航指令$\mathcal{W}$和视觉观察$\mathrm{R}_t$从可导航的候选移动方向中选择一个移动到达相邻的其它节点。

Motivation



(a) *Pipeline*

Table 1: Performance (on R2R validation unseen split) on different datasets solely. Directly training with R2R yields the best SPL compared to training with other augmentation datasets.

Training Data	#data	#Env.	SR↑	SPL↑
R2R	14K	61	65.9	**55.9**
Prevalent	1.0M	60	**67.1**	54.8
ScaleVLN	4.9M	800	63.9	50.1

- The process iterates, with the navigator improving the instruction generator via data filtering;
- And the instruction generator improving the navigator via data refining;
- Ultimately producing both a highly capable navigator and instruction generator, along with a substantially high-quality synthetic VLN dataset.

[Model Input]
This is a navigation trajectory consists of several image sequences:
(Viewpoint 1: Image:<image>, Action: left (135.71 degree) and up (0.00 degree)),
(Viewpoint 1: Image:<image>, Action: forward),
(Viewpoint 2: Image:<image>, Action: left (33.14 degree) and up (0.00 degree)),
...
(Viewpoint 5: Image:<image>, Action: forward),
(Viewpoint 6: Image:<image>, Action: stop).
Could you give me its corresponding navigation instruction in details?

[Model Output]
Start by exiting the bedroom into the hallway. Turn right and proceed down the hall until you reach the first door on your left. Enter this room, which is the library. Once inside, walk towards the large bookshelf directly across from the doorway. Stop there and face towards the window on the right side of the room.

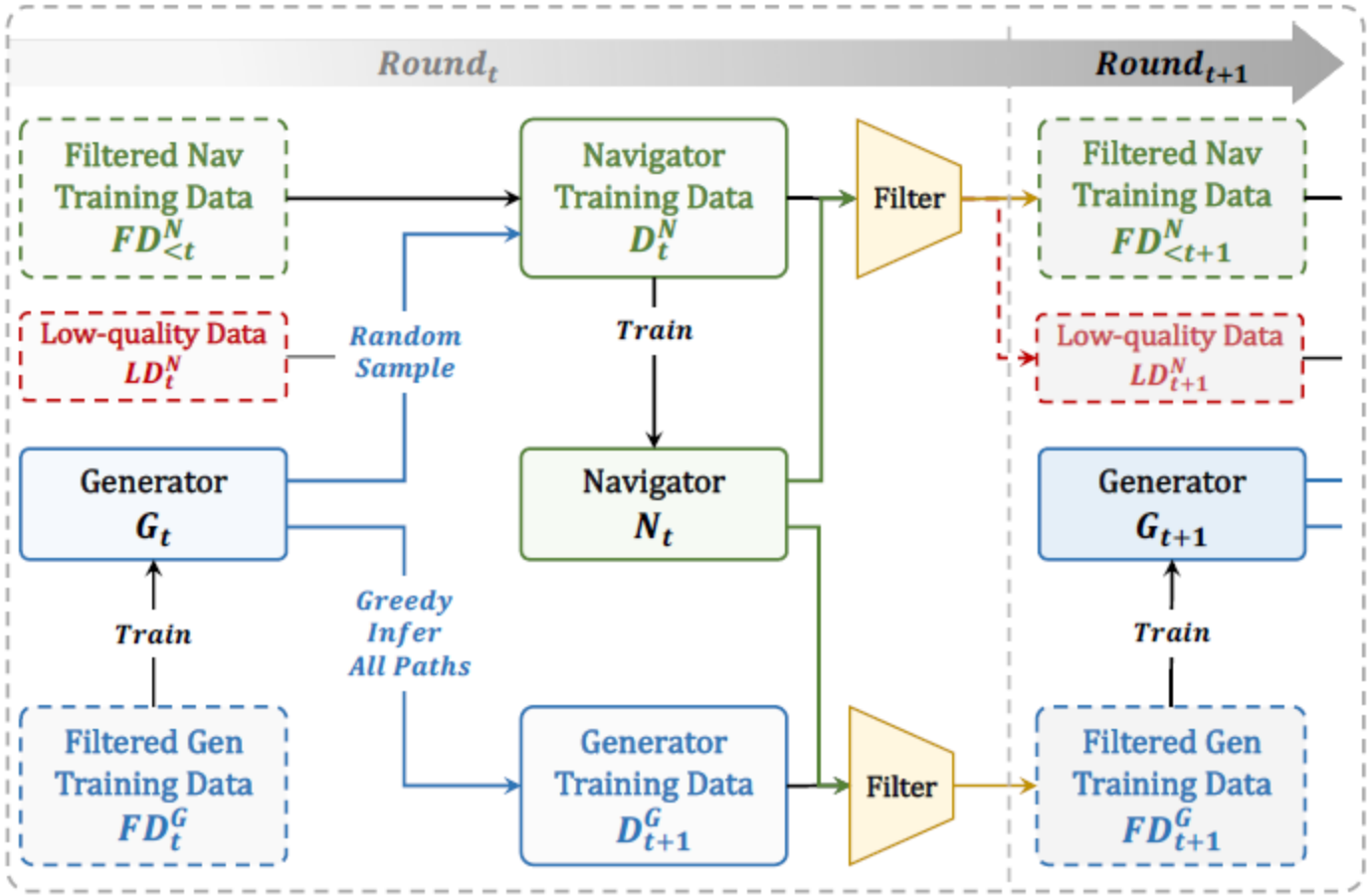
Figure 3: Interleaved trajectory prompt and expected response for training our instruction generator. The LLM's response is the corresponding instruction of this path.

Algorithm 1 Pipeline of Self-Refining Data Flywheel (SRDF)

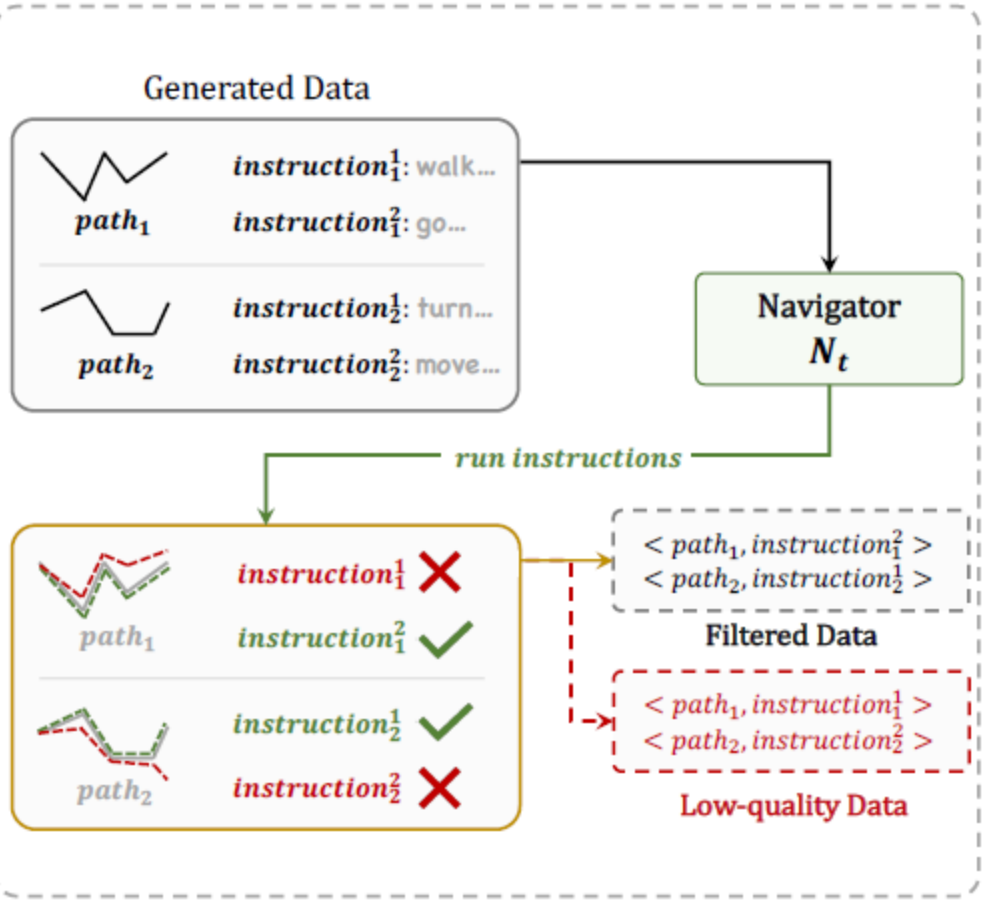
Require: Seed data D_{Seed} (Human-annotated), Unlabelled trajectories D_{Traj}, Total iterations T.

Train base instruction generator G_1 with D_{Seed}.
Use G_1 to generate nav-training data D_1^N and gen-training data D_2^G for D_{Traj}.
/* D_1^N *is generated via random sampling while* D_1^G *via greedy decoding*
Train base navigator N_1 with D_1^N.
Use N_1 to filter high-quality subsets FD_2^G from D_2^G and $FD_{<2}^N$ from D_1^N.
for each iteration t $(1 < t \leqslant T)$ **do**
    /* *Note: Seed data* D_{Seed} *is used in training stages of both* G_t *and* N_t *but omitted for simplicity*
    Train generator G_t with FD_t^G.
    Use G_t to generate nav-training data ND_t^N for LD_t^N and gen-training data D_{t+1}^G for D_{Traj}.
    /* ND_t^N *is generated via random sampling while* D_{t+1}^G *via greedy decoding*
    Combine ND_t^N and $FD_{<t}^N$ to form D_t^N.
    Train navigator N_t with D_t^N.
    Use N_t to filter high-quality subsets FD_{t+1}^G from D_{t+1}^G and FND_t^N from ND_t^N.
    Combine FND_t^N and $FD_{<t}^N$ to form $FD_{<t+1}^N$.
end for

Method



(a) Single-round Pipeline



(b) Filter Pipeline

Experiments

Table 7: Comparison of single-run performance on R2R and R2R-CE datasets. Best results are marked in **bold blue** and second best in **bold**.

Methods	Room-to-Room Dataset								Room-to-Room-CE Dataset					
	Validation-Unseen				Test-Unseen				Validation-Unseen			Test-Unseen		
	NE↓	OSR↑	SR↑	SPL↑	NE↓	OSR↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	NE↓	SR↑	SPL↑
Human	-	-	-	-	1.61	90	86	76	-	-	-	-	-	-
Seq2Seq (Anderson et al., 2018b)	7.81	28	21	-	7.85	27	20	-	-	-	-	-	-	-
Speaker Follower (Fried et al., 2018)	6.62	45	36	-	6.62	-	35	28	-	-	-	-	-	-
RCM (Wang et al., 2019)	6.09	50	43	-	6.12	50	43	38	-	-	-	-	-	-
EnvDrop (Tan et al., 2019)	5.22	-	52	48	5.23	59	51	47	-	-	-	-	-	-
PREVALENT (Hao et al., 2020)	4.71	-	58	53	5.30	61	54	51	-	-	-	-	-	-
NvEM (An et al., 2021)	4.27	-	60	55	4.37	-	58	54	-	-	-	-	-	-
AirBert (Guhur et al., 2021)	4.10	-	62	56	4.13	-	62	57	-	-	-	-	-	-
VLN↻BERT (Hong et al., 2021)	3.93	-	63	57	4.09	70	63	57	-	-	-	-	-	-
HAMT (Chen et al., 2021)	2.29	-	66	61	3.93	72	65	60	-	-	-	-	-	-
HOP (Qiao et al., 2022)	3.80	-	64	57	3.83	-	64	59	-	-	-	-	-	-
HOP+ (Qiao et al., 2023a)	3.49	-	67	61	3.71	-	66	60	-	-	-	-	-	-
DUET (Chen et al., 2022b)	3.31	81	72	60	3.65	76	69	59	-	-	-	-	-	-
Lily (Lin et al., 2023)	2.90	-	74	62	3.44	-	72	60	-	-	-	-	-	-
DreamWalker (Wang et al., 2023a)	-	-	-	-	-	-	-	-	5.53	49	44	5.48	49	44
BEVBert (An et al., 2023a)	2.81	84	75	64	3.13	81	73	62	4.57	59	50	**4.70**	**59**	**50**
ScaleVLN (Wang et al., 2023e)	**2.09**	**88**	**81**	**70**	**2.27**	**86**	**80**	**70**	4.80	55	51	5.11	55	50
GridMM (Wang et al., 2023d)	2.83	-	75	64	3.13	-	73	62	5.11	49	41	5.64	46	39
ETPNav (An et al., 2023b)	-	-	-	-	-	-	-	-	4.71	57	49	5.12	55	48
DualAction (Zhang & Kordjamshidi, 2024)	-	-	-	-	-	-	-	-	-	58	49	-	56	48
HNR (Wang et al., 2024e)	-	-	-	-	-	-	-	-	**4.42**	**61**	**51**	4.81	58	50
NaviLLM (Zheng et al., 2024)	3.51	-	67	59	3.71	-	68	60	-	-	-	-	-	-
NavGPT-2 (Zhou et al., 2024a)	2.84	84	74	61	3.33	80	72	60	-	-	-	-	-	-
VER (Liu et al., 2024)	2.80	-	76	65	2.74	-	76	66	-	-	-	-	-	-
MAGIC-L (Wang et al., 2024b)	2.22	86	79	**70**	2.75	82	77	69	-	-	-	-	-	-
GOAT (Wang et al., 2024a)	2.40	85	78	68	3.04	80	75	65	-	-	-	-	-	-
SRDF (Ours)	**1.62**	**90**	**86**	**79**	**1.82**	**89**	**85**	**78**	**4.12**	**65**	**57**	**4.35**	**64**	**56**

Experiments

Table 2: Statistics of training data on different VLN datasets.

Dataset	Instruction	#Env.	#Instr.	#Vocab.	Instr. Length
R2R (Anderson et al., 2018b)	Manually Labelled	61	14,039	3,063	26.33
RxR-en (Ku et al., 2020)		60	26,464	7,249	102.13
REVERIE (Qi et al., 2020)		60	10,466	1,140	18.64
CVDN (Thomason et al., 2020)		57	4,742	2,068	53.21
SOON (Zhu et al., 2021)		34	2,780	735	44.09
R4R (Zhu et al., 2020)		59	233,532	3,004	52.25
Prevalent (Hao et al., 2020)	Generated	60	1,069,620	993	24.23
Marky (Wang et al., 2022b)		60	333,777	2,231	99.45
AutoVLN (Chen et al., 2022c)		900	217,703	1,696	20.52
ScaleVLN (Wang et al., 2023e)		1289	4,941,710	172	21.61
SRDF-20M (Ours)		860	20,417,874	10,363	24.05

Table 3: Navigator and instruction generator results in different rounds in R2R val unseen split.

Method	Instruction Following				Instruction Generation						
	NE↓	OSR↑	SR↑	SPL↑	SPICE↑	SPICE-D↑	Bleu-1↑	Bleu-4↑	CIDEr↑	Meteor↑	Rouge↑
Baseline	2.37	85.5	78.6	69.9	21.8	28.0	72.5	27.7	42.2	23.6	49.0
Ours (round 1)	1.95	87.1	82.4	75.9	23.7	28.4	71.4	29.5	46.5	23.1	50.2
Ours (round 2)	1.81	88.5	83.6	77.3	25.2	29.9	73.7	**31.0**	**50.7**	24.2	**51.3**
Ours (round 3)	**1.76**	**89.6**	**84.4**	**77.6**	**25.7**	**30.4**	**74.5**	30.8	49.7	**24.5**	**51.3**

NavRAG: Generating User Demand Instructions for Embodied Navigation through Retrieval-Augmented LLM

NavRAG: Generating User Demand Instructions for Embodied Navigation through Retrieval-Augmented LLM

Zihan Wang[2], Yaohui Zhu[3], Gim Hee Lee[2], Yachun Fan[1] ✉
[1]School of Artificial Intelligence, Beijing Normal University
[2]School of Computing, National University of Singapore
[3]College of Information Science and Technology, Beijing University of Chemical Technology
zihan.wang@u.nus.edu, fanyachun@bnu.edu.cn

ACL 2025 Findings

2025/10/17

Motivation

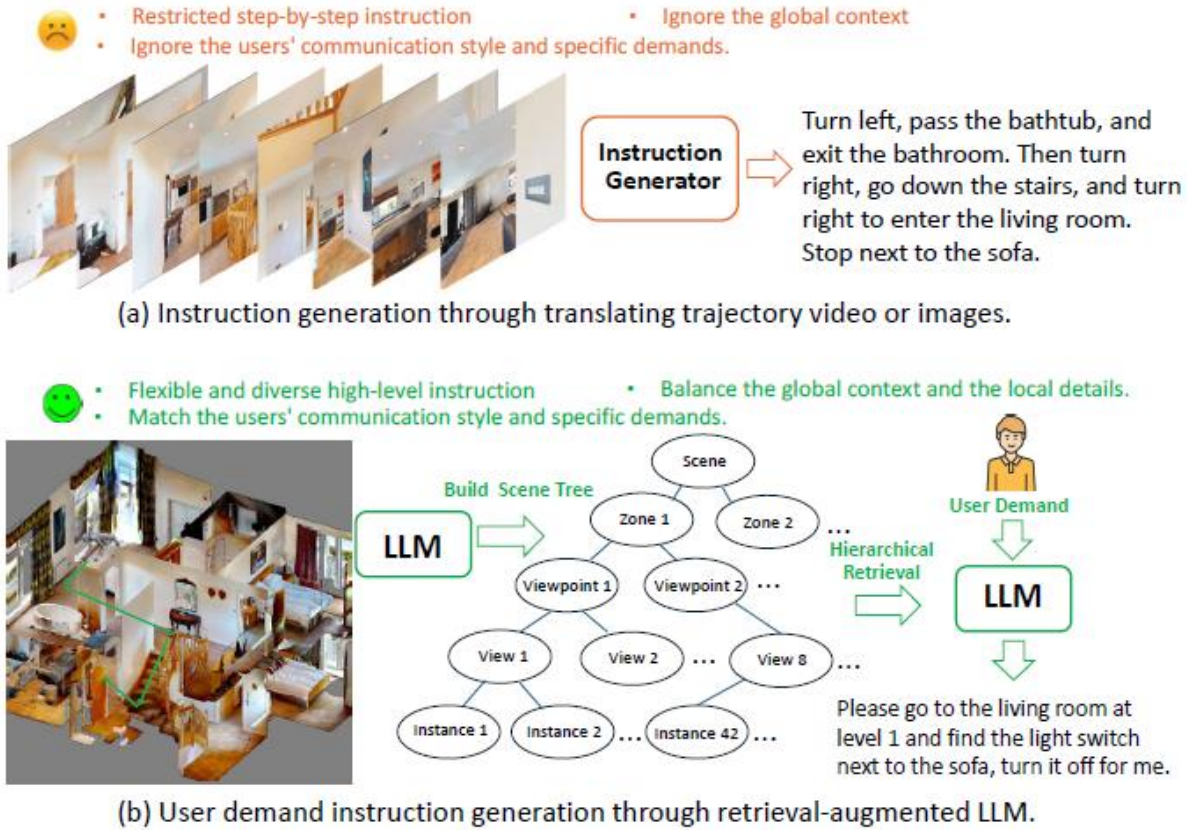


Figure 1: The comparison of previous navigation instruction generation methods (a) and NavRAG (b).

- This work proposes an approach for automatically constructing scene description trees and generating user demand navigation instructions using retrieval-augmented LLM.
- We annotate over 2 million high-quality navigation instructions across 861 3D scenes for training and evaluation.
- The VLN models trained on our NavRAG dataset achieve superior performance on VLN benchmarks, validating the effectiveness of the proposed method.

Method

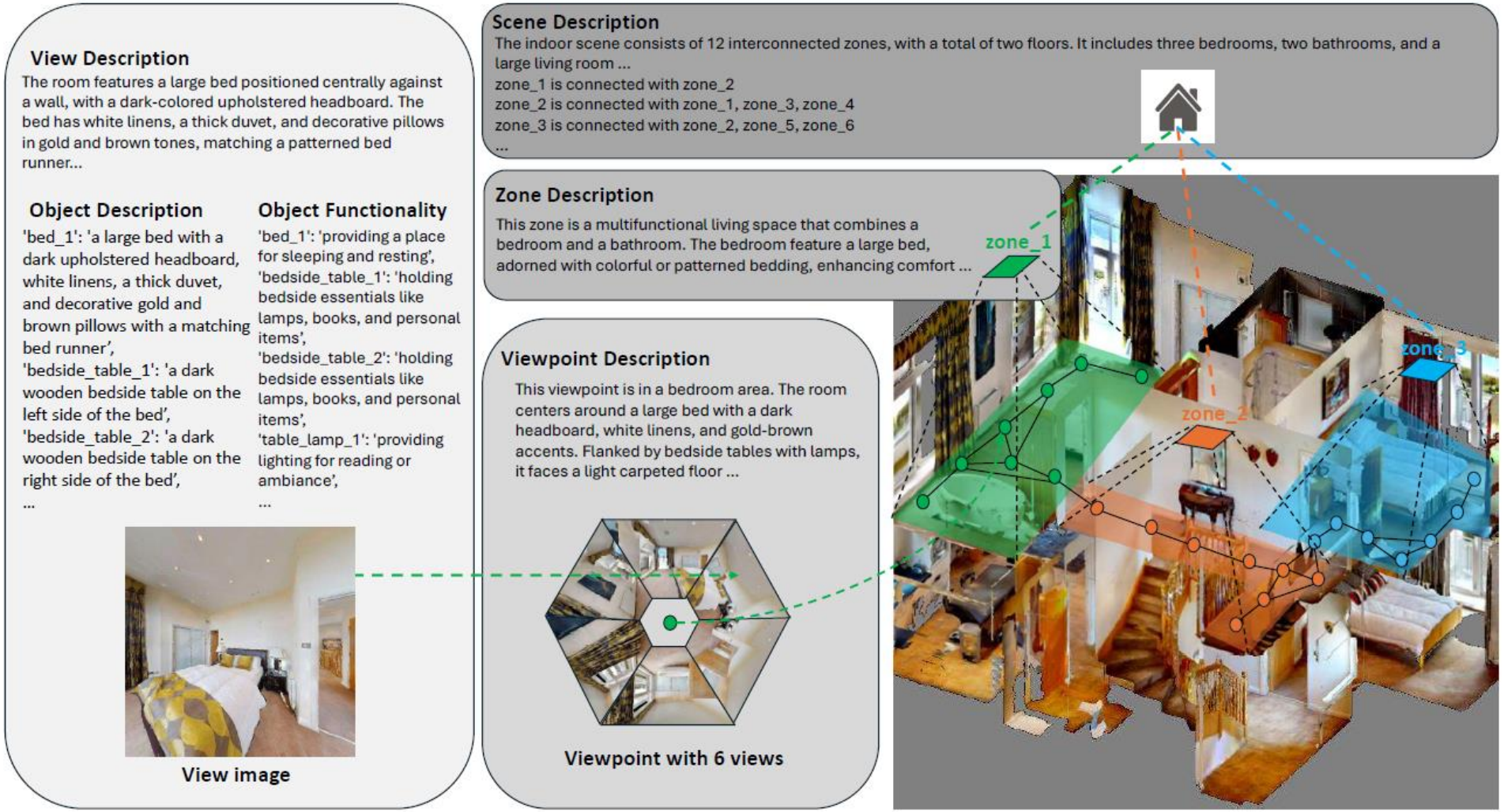


Figure 2: Demonstration of the Scene Description Tree. Based on LLM, NavRAG constructs the scene description tree in a bottom-up manner, progressively constructing from objects to views, viewpoints, zones, and the overall scene. This hierarchical structure describes environmental semantics and spatial relationships at different levels, facilitating LLM in understanding 3D environments and retrieving information for instruction generation.

Method

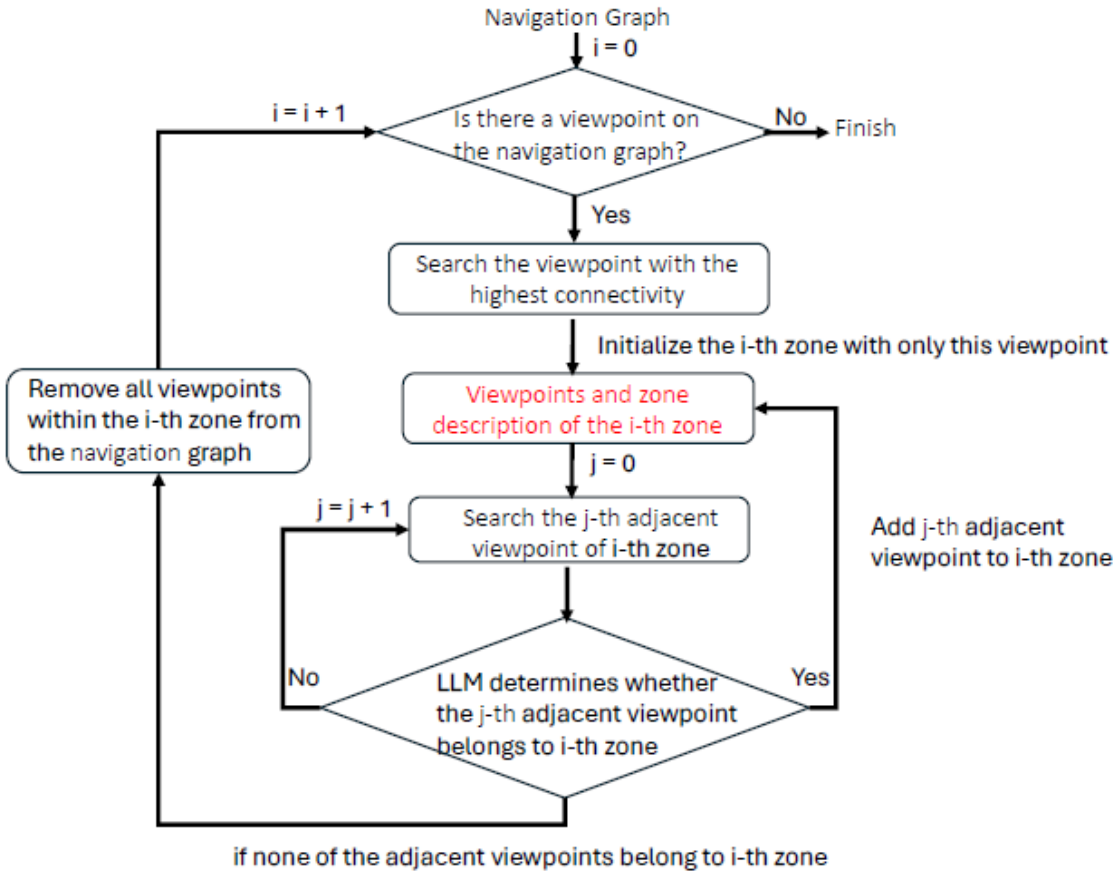


Figure 3: Framework of the zone partitioning algorithm based on connectivity relations and environmental semantics.

Method

{
"Age": 33,
"Gender": "Female",
"Occupation": "Lawyer",
"Lifestyle Description": "You maintain the good habit of going to bed early and waking up early. Besides working in the study, you often do yoga and other exercises in the living room and enjoy cooking your own meals."
}

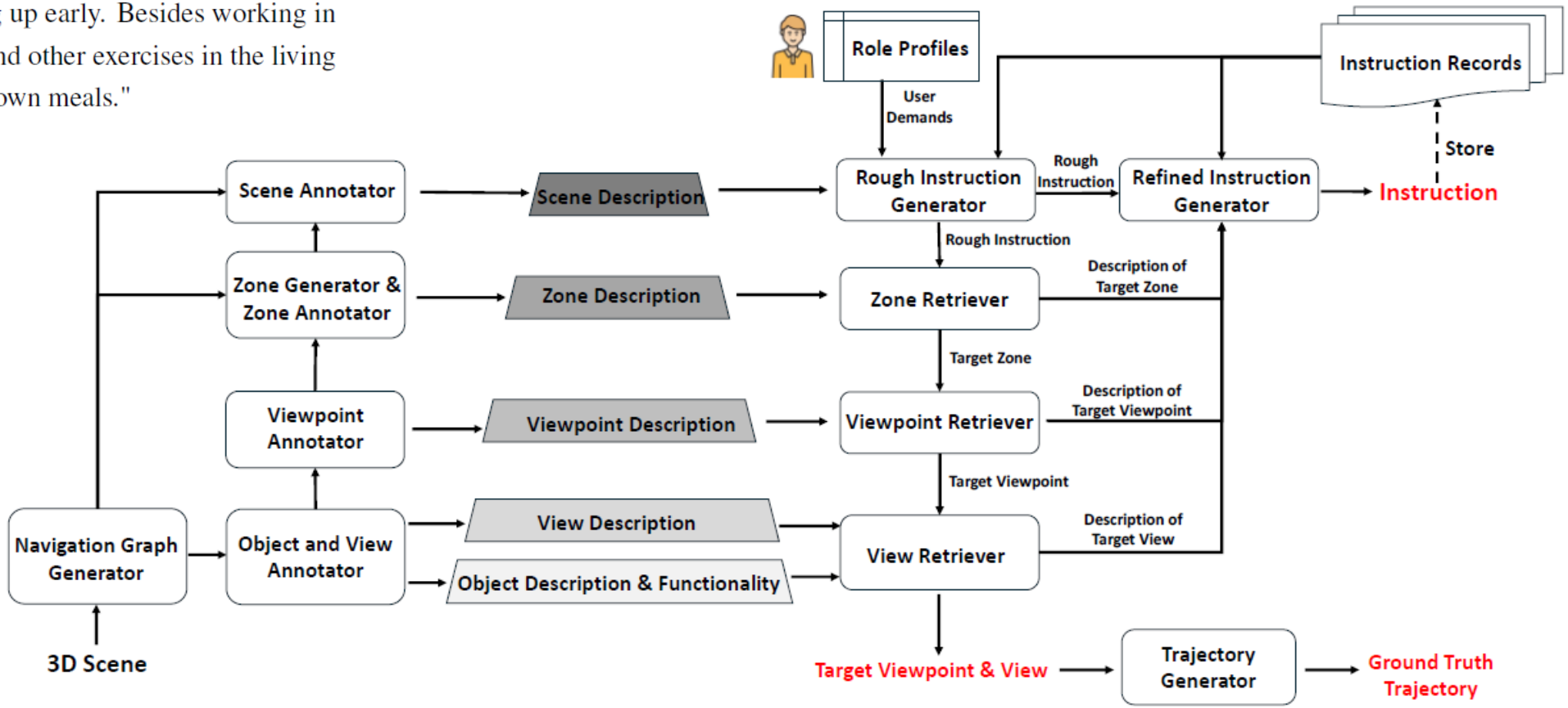


Figure 4: Framework of NavRAG for scene tree construction and navigation instruction generation through Retrieval-Augmented LLM.

Method

Rough Instruction Generation

Prompt Now, you are a user of a household assistant robot, and your personal information is as follows:

Input {"Age": 28, "Gender": "Female", "Occupation": "Graphic Designer", "Lifestyle Description": "You are a creative individual who enjoys working from home. You spend long hours on your computer designing, and your work environment needs to be tidy and inspiring. You rely on the robot to handle cleaning and organizing your space while you focus on your projects."}

Prompt Your house is divided into several zones, with each zone's connectivity and environmental description provided below:

Input
zone_1 is connected with zone_5, zone_6, zone_11, zone_12, zone_13, zone_15, zone_18, zone_19
zone_2 is connected with zone_9, zone_14, zone_17
...
{ "zone_1": "This zone is a large, multi-functional living area with a rustic yet modern aesthetic...Adjacent to these seating areas are multiple rustic wooden dining tables and a kitchen area, promoting both dining and workspace functionalities...",
"zone_2": "This zone is a rustic multi-functional living area and hallway...", ... }

Prompt Please simulate the user and record your instructions to the household assistant robot throughout your day, from waking up to going to bed. Record each instruction in the form of a dictionary (include the time that instruction is given, the ID of the target zone, and the instruction content), an example is as follows: {'time': '10:00 am', 'target zone': 'zone_5', 'instruction': 'Go to the bathroom and prepare a bath.'}. Do not include multiple tasks in one instruction. Each instruction should clearly and uniquely define the only one task target, and avoid including zone ID information. Here are some existing instruction records:

Input {'time': '9:00 am', 'target zone': 'zone_5', 'instruction': 'Please head to the stylish bathroom featuring a round vessel sink; make sure to wipe down the stone countertop and tidy up any items left around the space.'}
{'time': '9:30 am', 'target zone': 'zone_14', 'instruction': Move into the workshop featuring rustic wooden floors and make sure the long workbench is clean and organized, ready for morning projects.'}
{'time': '10:40 am', 'target zone': 'zone_13', 'instruction': 'Head to the rustic loft with a large, soft sofa and arrange the decorative cushions for added comfort.'}

Prompt Please output the next instruction record, don't output anything other than a correct dictionary.

Output {'time': '11:00 am', 'target zone': 'zone_1', 'instruction': ' Walk to the warm hall and set the wooden table for lunch.'}

Method

Hierarchical Retrieval

Prompt These are the environmental descriptions of some viewpoints within zone_1 (recorded as a dictionary, with keys as viewpoint_id and values as descriptions). Which viewpoint is the most likely location of the robot's task target? Please output the string of the viewpoint_id, ensure the viewpoint_id is correct and in the keys of dictionary and don't output anything other than it:

Input { "viewpoint_1": "This viewpoint is in a rustic interior space that features a combination of a hallway and living area... ",
"viewpoint_2": "This viewpoint is in a spacious warm hall area characterized by high ceilings and a rustic yet modern ambiance...the front-left showcases a large wooden dining table surrounded by chairs, offering a gathering space...",
"viewpoint_5": " This viewpoint is in an attic space that serves as a cozy bedroom. A large bed with layered bedding is situated against the left wall ... ",
...}

Output "viewpoint_2"

Prompt The following are text descriptions observed from the selected viewpoint_2 in six directions (forward, front-right, back-right, backward, back-left, and front-left), recorded as a dictionary, with keys as direction IDs and values as descriptions. Which direction is most likely to contain the robot's task target? Please output the string of the direction ID, and don't output anything other than it:

Input { "forward" :
...
"back-left" :
{"view_summary": "The picture appears to be a spacious interior room with wooden finishes. There is a large wooden dining table in the center of the room, with candles and napkins. The room features large windows on one side, providing ample natural light. To the left of the dining table, there is a stairway leading to an upper level. The area has a cozy ambiance due to the wooden beams and flooring.", "instance_description": {"dining_table_1": "a large rectangular wooden dining table situated in the center", "staircase_1": "a wooden staircase to the left of the dining table"}, "instance_affordance": {"dining_table_1": "used for dining and gathering; placing items for meals", "staircase_1": "providing access to an upper level of the interior space"}},
... }

Output "back-left"

Figure 5: Prompt, input and output of the Rough Instruction Generator and Hierarchical Retrieval.

Rough Instruction Generation

Refined Instruction Generation

Input The description of the target zone is: $ZONE. The description of the target viewpoint is: $VIEWPOINT. The description of the target direction within this viewpoint is: $VIEW. The main instances within this direction are: $INSTANCE, and their functions are: $AFFORDANCE. Please modify your instruction to ensure it is accurate and clearly specifies only one destination (i.e., does not exist in other zones and different from the destination in previous instructions). You can modify the description of it, use more diverse objects and spatial relations, use more varied range of sentence structures for better diversity. An example is as follows: {'time': '10:00 am', 'target zone': 'zone_5', 'instruction': 'Go to the bathroom with a large mirror and fill the white bathtub with hot water for me.'}. The instructions should be phrased in a natural, concise, conversational tone. Don't output anything other than a correct dictionary, don't use code blocks in Markdown.

Output {'time': '11:00 am', 'target zone': 'zone_1', 'instruction': 'Walk to the warm hall featuring elegant wooden accents and set the large wooden table with candles and napkins for a lovely dinner ambiance.'}

Figure 6: Prompt, input and output of the Refined Instruction Generator. $ZONE, $VIEWPOINT, $VIEW, $INSTANCE and $AFFORDANCE denote retrieved environmental descriptions at different levels.